Limits of Pixel-Space Anomaly Detection for Deepfakes: Implications for Feature-Space and Foundation Model Approaches

Nipun Joisa
Dept. of CSE - AI & ML
RNS Institute of Technology
Bengaluru, India

***Abstract*—Deepfakes have become so realistic that it has become challenging to differentiate authentic media from manipulated media. As technology keeps advancing, we need systems that can detect futuristic deepfake content . Supervised deep-learning models are widely used today, but they tend to break down when they're exposed to new types of deepfakes they weren't trained on.In this study, we explore whether anomaly-detection models trained only on authentic facial images can generalise more effectively to unseen deepfake methods. We compare a supervised XceptionNet classifier with two unsupervised alternatives: a standard Autoencoder and a Variational Autoencoder (VAE). All three models were trained and evaluated using a combined dataset built from FaceForensics++ and Celeb-DF.**

Our results show that the supervised classifier performs well, achieving an F1 Score of 0.89 for detecting fake images. In contrast, both anomaly-based models operate close to chance level, with accuracies around 50%. These outcomes suggest that, for today's high-quality, diverse deepfakes, supervised learning on a large, varied dataset remains the more reliable approach. They also indicate that learning only the distribution of real faces is insufficient for detecting the subtle artefacts in modern deepfake generation.

This study quantifies how modern deepfakes affect pixel space anomaly detection and provides empirical evidence supporting the field's shift toward feature space and foundation model based approaches.

***Index Terms*—Deepfake Detection, Anomaly Detection, Supervised Learning, Autoencoder, Variational Autoencoder (VAE), Computer Vision, Generalisation**

I. Introduction

Modern AI has caused deepfake videos to evolve incredibly fast. Tools like GANs, diffusion models, and neural rendering are capable of producing fake footage that looks real [1]. Of course, this tech has valid uses in movies or for avatars, but the risks are high.We are seeing it used more often for identity theft, fake news, and non-consensual media [2], [3].Since most people actually struggle to tell the difference between a real video and a deepfake [4], we can't just rely on the naked eye anymore, we need reliable software to catch them.Most existing deepfake detectors use supervised deep learning, especially CNN-based models trained on labelled real and fake images [5], [6].These models usually perform well on the datasets they were trained on. However, a major issue is that the supervised models fail to generalise to new or unseen deepfakes. [7], [8]. This challenge is only getting harder as generation tools improve, driven by new datasets such as DFDC [9], KoDF [10], and DeepSpeak [11].Because the field keeps evolving rapidly, the older techniques seem to get outdated and inefficient.

Because of this, the research focus is shifting.We need approaches that work on unseen deepfakes, not just the ones we already know. One idea is to train models strictly on real faces. The theory is simple: if the model learns the features of genuine footage perfectly, it should be able to spot anything 'unusual'—regardless of the specific trick used to create it. That hypothesis is the driving force behind the comparison in this paper.

In this study, we evaluate a supervised XceptionNet classifier alongside two anomaly-detection models: a standard Autoencoder and a Variational Autoencoder (VAE). To test their capabilities, we ran all three models on a dataset derived from both FaceForensics++ [12] and Celeb-DF [13]. Our goal is to understand whether anomaly detection can provide better generalisation than supervised learning.

II. Related Work

Deepfake detection has evolved quickly because manipulated videos are becoming more realistic every year. Researchers have developed many approaches to keep up with these improvements. This section discusses the methods most relevant to our study.

A. Supervised, Frame-Based Detection

A common way to detect deepfakes is to treat the task like a simple classification problem: each video frame is checked to decide if it is real or fake [2].CNN models, especially strong ones like XceptionNet [14], often achieve high accuracy on standard benchmark datasets [5].Several large datasets—such as DFDC [9], FaceForensics++ [12], Celeb-DF [13], and DeepFakeBench [15]—have supported these supervised detection methods.

However, supervised models often have a major limitation: they tend to fail when they encounter manipulation styles not present in their training data [7], [8]. This happens because the model often learns patterns specific to a dataset instead of learning real, general features [5]. Newer deepfake methods—such as diffusion-based identity swapping [16] or one-shot neural reenactment [1]—can easily fool detectors trained on older data.

B. Unsupervised and Anomaly-Based Detection

To overcome this generalisation gap, researchers have explored unsupervised approaches. These models learn how real facial images normally look and then mark unusual spots as suspicious [5]. Autoencoders and VAEs are often used for this purpose [2], [17]. Some early works showed us promising results [18], but as deepfakes continue to become more realistic, reconstruction based methods started failing because fakes are now as good as authentic ones.

More recent studies have explored better strategies, such as self supervised pretraining [19], contrastive learning [20], and latent space augmentation [21]. These methods aim to learn more general and manipulation independent features.

Table I summarises key deepfake detection methods from prior work, including their architectures, datasets, and reported performance.

C. Advanced and Temporal Approaches

Instead of just looking at static frames, many researchers are now trying to find glitches that happen over time. Sun et al. and Guo et al. use facial dynamics to identify motion that feels unnatural [29], [30]. Other studies are taking a similar route by targeting issues like bad lip syncing [31] or strange head angles [32].

At the same time, the field is advancing towards foundation models like CLIP. With their GenD method, Yermakov et al. [8] proved that you can get great results with just a little bit of tuning on a pre-trained model. We see the same trend in works like Forensics Adapter [33] and FaceGuard [34]. Together, these studies make a strong case that the future of detection isn't about training new models from scratch, but about adapting the big ones we already have.

III. Methodology

In this section, we demonstrate exactly how we prepared the data and trained the models. We provide all the details needed to make the process easy to follow and reproduce.

A. Dataset Construction

We used a dataset combined from FaceForensics++ [12] and Celeb DF [13]. These two datasets were selected because they contain a wide range of real and manipulated facial videos. FaceForensics++ provides several forgery methods such as DeepFakes, FaceSwap and Neural Textures. Celeb DF contains high quality and difficult examples that are closer to real world conditions.

From each video, frames were sampled at regular intervals in order to capture different facial expressions and lighting conditions. The dataset was balanced with an equal number of real and fake samples. Seventy percent of the data was used for training, fifteen percent for validation, and fifteen percent for testing. Only real samples were used when training the Autoencoder and the VAE, because the goal of these models is to learn the natural distribution of real faces.

To keep things fair, we balanced the data so there was an even mix of real and fake videos. We used a standard 70-15-15 split for training, validation, and testing. During training, we showed the Autoencoder and the VAE only real faces. This was necessary to support the anomaly-detection method we used.

B. Pre-processing Pipeline

Every video went through the same preprocessing steps. To prepare the data, we pulled a set number of frames from the videos. We used MTCNN to locate the face and crop it to 299 by 299 pixels. For the classifier model specifically, we applied random changes like zooming or flipping the image horizontally/vertically. This makes the model more robust. Overall, this workflow ensured that every model got clean data to work with.

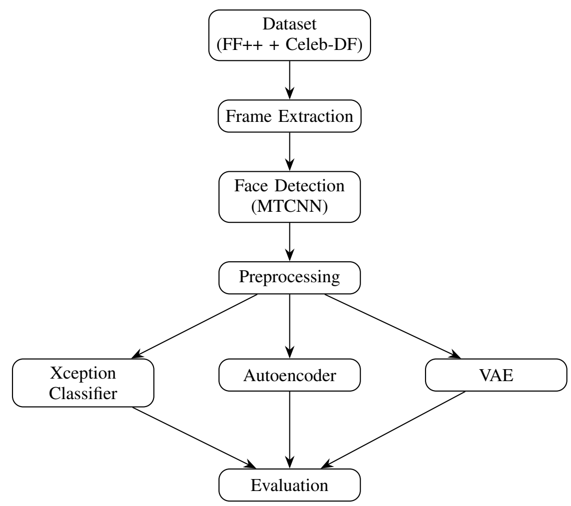

Fig. 1. Overall workflow from dataset to evaluation.

C. Model Architectures

The study compared three different models. One is a supervised classifier based on XceptionNet. The other two are anomaly detection models, an autoencoder and a variational autoencoder. Each model receives the same preprocessed face images, but they learn in different ways.

1) Model A: Supervised XceptionNet Classifier: The supervised model uses XceptionNet [14] as its feature extractor. The backbone was initialised with ImageNet weights so that the model already understood general visual patterns before training. The final layers of the backbone were fine tuned on the deepfake task. A custom classification head was added, which contained global average pooling, a dense layer with five hundred and twelve units, dropout of fifty percent and a sigmoid output layer.

Training followed a straightforward procedure. Each input image was passed through the backbone and classification head to obtain a prediction score. Binary cross entropy was used as the loss function. The model weights were updated after each batch using the Adam optimiser [17]. The validation

TABLE I
COMPARISON OF EXISTING DEEPFAKE DETECTION METHODS

Paper	**Method**	**Model**	**Dataset**	**Accuracy**	**AUC**
Amerini et al. [22]	Optical Flow + CNN	VGG16 / ResNet50	FF++	81.61%	-
			FF++	75.46%	-
Peng Chen et al. [23]	CNN + LSTM	VGG16	FF++	-	100
			Celeb-DF	-	77.6
			UADFV	-	91.1
			Deepfake TIMIT (HQ)	-	98.5
			Deepfake TIMIT (LQ)	-	99.5
Afchar et al. [6]	CNN	Meso-4	FF++ (Face2Face)	95%	-
		MesoInception-4	FF++ (Face2Face)	98%	-
Aneja et al. [24]	CNN + LSTM	ResNet18	FF++	92.23%	-
			Google DFD	81.21%	-
			AIF	60.79%	-
			Dessa	74.28%	-
			Celeb-DF	68.83%	-
Ranjan et al. [25]	CNN + LSTM	XceptionNet	Celeb-DF	83.49%	-
			DFDC	78.13%	-
			DFDC	94.33%	-
Li et al. [26]	Multiple Instance Learning	XceptionNet	Celeb-DF	85.11%	-
			DFDC	98.84%	-
			FFPMS	90.71%	-
De Lima et al. [27]	Spatio-temporal CNN	RCN	Celeb-DF	76.25%	74.87
		R2Plus1D	Celeb-DF	98.07%	99.43
		I3D	Celeb-DF	92.28%	97.59
		R3D	Celeb-DF	98.26%	99.73
		MC3	Celeb-DF	97.49%	99.30
Khan et al. [28]	CNN	VGG16	DFDC	96.75%	-

set was used to monitor overfitting and to decide when to stop training.This model learns directly from both real and fake samples and attempts to classify them based on discriminative features present in each image.

We included a block diagram to visualize how the model processes data. It shows the journey starting from the input frame, moving through the Xception backbone and classification head, and ending with the final result.

The supervised classifier is trained using the binary cross entropy loss. For a given input image with label $y \in \{0, 1\}$ and predicted probability $\hat{y}$, the loss is defined as:

$$\mathcal{L}_{\text{BCE}} = -\left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})\right]. \quad (1)$$

The Adam optimiser is used to update the parameters during training.

2) Visualisation of Model Predictions: Figure 3 provides a qualitative analysis of the model behaviour. It shows that the classifier tends to focus on key facial regions such as the eyes, nose, and mouth, where manipulation artefacts are more likely to appear. The predicted regions closely align with the ground truth in most cases, indicating that the model is learning meaningful visual features rather than relying on random patterns.

3) Model B: Autoencoder: Our autoencoder uses a basic encoder and decoder architecture. The encoder takes the input image and uses convolution layers to squeeze it down. This creates a compressed version of the data known as a latent representation. The decoder uses transpose convolution layers to reconstruct the original image from this latent code.

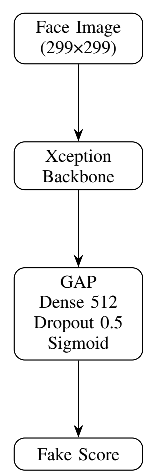

Fig. 2. Pipeline for the supervised Xception classifier. GAP denotes global average pooling.

The autoencoder sees only real faces during training. It attempts to recreate these images as accurately as it can by reducing the mean squared error. When we test a new image, the reconstruction error acts as an alert. A high error means the model is having trouble rebuilding the image. We use a specific threshold value to make the final call on whether it is fake or real.

The training process can be described simply. An input image is encoded, decoded and compared with the original. The loss is calculated and the encoder and decoder weights are updated. A diagram for this model can show how the encoder compresses the image, how the bottleneck is formed and how

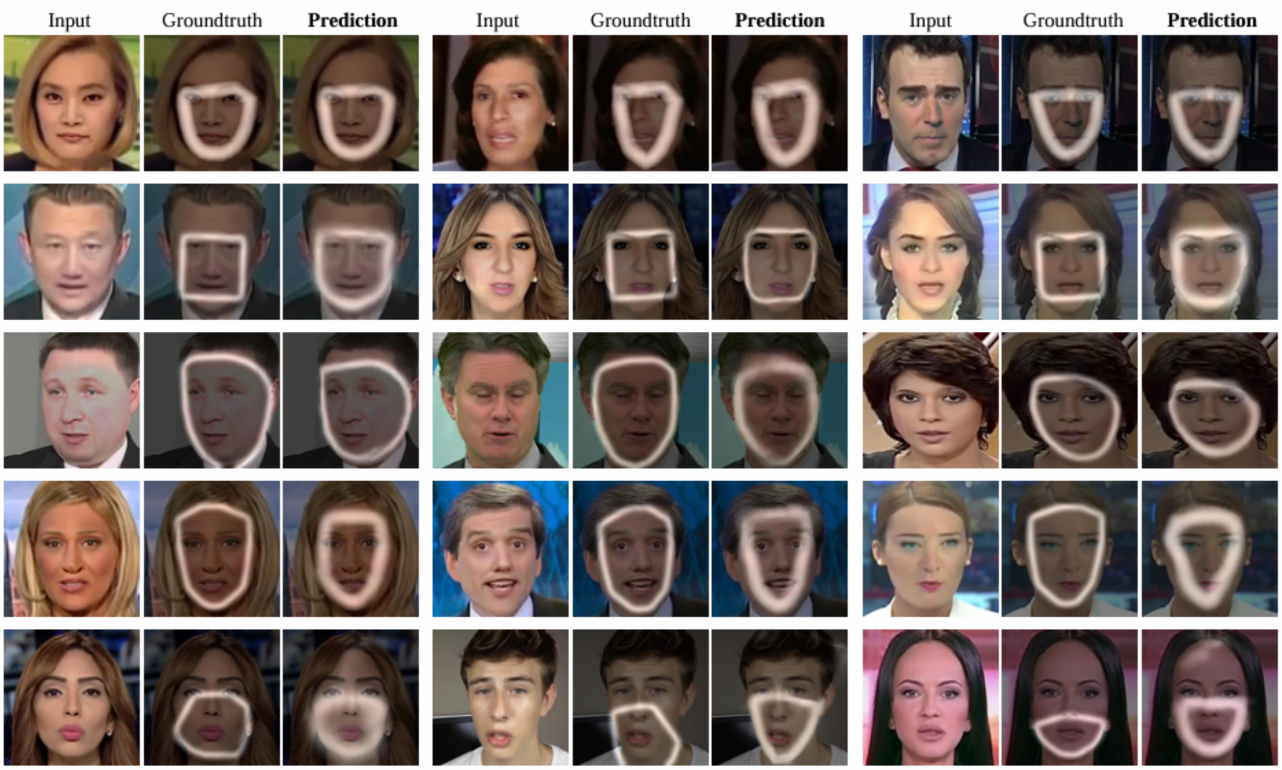

Fig. 3. Visual comparison between input images, ground truth regions, and model predictions. The highlighted regions indicate areas where the model focuses while identifying manipulated facial features.

the decoder rebuilds the output.

The autoencoder is trained only on real face images using the mean squared error between the input image x and its reconstruction $\hat{x}$. The loss is given by:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2. \tag{2}$$

A high reconstruction error during testing indicates that the model is unable to reproduce the input faithfully, which may suggest the presence of manipulation.

4) Model C: Variational Autoencoder (VAE): The VAE also uses an encoder and a decoder, but it learns a smoother and more structured latent space. The encoder does not produce a single latent vector. Instead, it outputs two vectors which represent the mean and the variance of a distribution.A sample is drawn from this distribution using the reparameterisation trick [17]. This sampled vector is then passed to the decoder which reconstructs the image.

The training process involves two parts, one is the reconstruction loss, which checks how close the copy is to the original. The other is Kullback Leibler divergence, which forces the data to fit a standard normal distribution. By combining these, the VAE learns a structured and continuous map of real faces.

For anomaly detection, the VAE relies on the idea that fake images should fall outside the learned distribution. If the reconstruction or the latent behaviour is unusual, the model flags the image as fake. A diagram for this model should include the encoder, the mean and variance branches, the sampling step and the decoder.

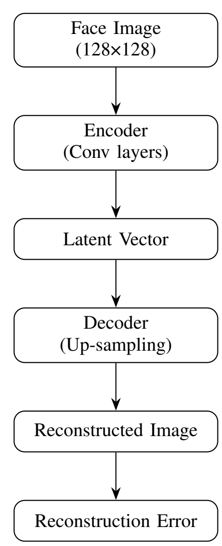

Fig. 4. Downward-flow pipeline for the Autoencoder anomaly detector.

The VAE uses the same reconstruction objective as the autoencoder, which is

$$\mathcal{L}_{\text{rec}} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2. \tag{3}$$

The encoder outputs a mean vector μ and a variance vector σ^2. The Kullback–Leibler divergence regularises the latent distribution as

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2}\sum_{j=1}^{d}\left(1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2\right), \tag{4}$$

where d is the dimension of the latent vector.

The full training objective for the VAE combines the reconstruction loss and the KL divergence. The evidence lower bound (ELBO) is defined as:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}. \tag{5}$$

Minimising this objective encourages accurate reconstructions while maintaining a smooth and structured latent space.

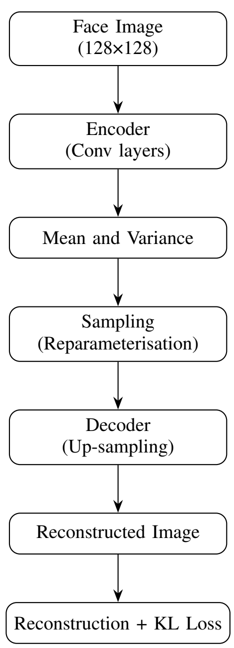

Fig. 5. Downward-flow pipeline for the Variational Autoencoder.

D. Overall Workflow

The entire system can be visualised through a single flow diagram. It begins with the combined dataset. It continues to frame extraction, face detection, preprocessing and normalisation. After this, the processed images are fed into three parallel paths: the supervised classifier, the autoencoder and the VAE. Their outputs are then compared during evaluation. This layout provides a clear picture of how the study was conducted from start to finish.

IV. Results

This section explains how the three models performed on the test set of 4,440 face images. The test data contained equal number of real and fake images. The evaluation focuses on accuracy, precision, recall, and the F1 score to assess how well each model distinguishes real from fake images.

A. Overall Performance Comparison

Among the three models, the supervised XceptionNet classifier gave the best results. As shown in Table II, it had an accuracy of 88.13 percent and an F1 score of 0.89 for the fake class. These results show that the classifier was able to generalise fairly well to the types of manipulations present in FaceForensics++ and Celeb DF.

The autoencoder and the VAE performed close to chance level. Since both models were trained only on real faces, they were not able to reliably separate real images from manipulated ones. The results show that reconstruction based detection is not effective for modern deepfakes, which often contain very subtle visual changes that are difficult to detect through reconstruction alone.

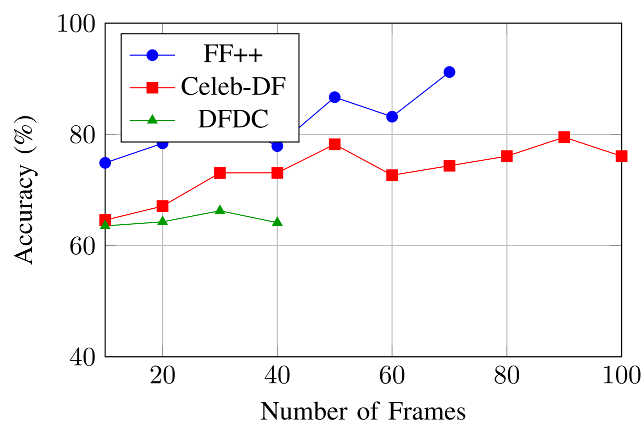

Fig. 6. Impact of number of frames on detection accuracy across FF++, Celeb-DF, and DFDC datasets.

B. Model A: Supervised Classifier (XceptionNet)

The supervised classifier showed strong and consistent performance. The overall accuracy reached 88.13 percent. For the fake class, the precision was 0.86, the recall was 0.91, and the F1 score was 0.89. These values show that the model detected most fake images correctly.

C. Model B: Autoencoder Anomaly Detector

The autoencoder was unable to distinguish real from fake images. During testing, the reconstruction errors for real and fake images were almost identical. As a result, the model labelled nearly every image as real. The overall accuracy dropped to 50 percent, which is no better than random guessing. The F1 score for the fake class was zero because the model did not identify any fake images correctly.

D. Model C: Variational Autoencoder (VAE)

The VAE showed a different type of failure. Although a threshold was found during validation, the model labelled almost every test image as fake. This produced a recall of 1.00 for the fake class, since it correctly marked all manipulated images. However, it also misclassified many real images. The precision was 0.50, and the overall accuracy was 50.18 percent. The F1 score was 0.67, showing that the model had high recall but poor balance between correct and incorrect predictions.

E. ROC Curve Analysis

To further evaluate the performance of the models, we analyse the Receiver Operating Characteristic (ROC) curves across different datasets. These curves illustrate the trade-off between true positive rate and false positive rate at different thresholds.

Figure 10 shows that the classifier achieves strong separability on FF++, with curves close to the top-left corner. Performance is slightly lower on Celeb-DF and further reduced on DFDC, reflecting the increased realism and difficulty of these datasets.

F. Comparison and Interpretation

Figure 11 shows a visual comparison of accuracy and F1 scores. From these results, it is clear that the supervised classifier was the only model that provided dependable performance. Both the autoencoder and the VAE struggled to tell real and fake images apart when the visual differences were very small. The VAE made use of its structured latent space, but this was still not enough for reliable detection.

Overall, the results suggest that reconstruction based anomaly detection is not well suited for the deepfakes produced today. These videos are generated by powerful tools that leave almost no visible marks. The supervised classifier has the upper hand here because it learns to spot the differences between real and fake images directly from the data. As long as you have a large, diverse dataset, supervised learning is still the most effective way to catch deepfakes.

TABLE II
COMPARATIVE PERFORMANCE OF DETECTION MODELS

Model	Accuracy	Precision	Recall	F1-Score
Model A: Classifier	88.13%	0.86	0.91	0.89
Model B: Autoencoder	50.00%	0.00	0.00	0.00
Model C: VAE	50.18%	0.50	1.00	0.67

V. DISCUSSION

The results in Table II and Figure 5 shows a clear gap in performance between the supervised classifier and the two anomaly detection models we used. These findings helps us understand the strengths and weaknesses of each approach and what they mean for modern deepfake detection.

A. The Success of Supervised Learning

The XceptionNet classifier came out on top with an F1 score of 0.89 for the fake category. This result confirms that the model can learn the specific details that distinguish real faces from fake ones. Although we saw signs of overfitting at the start, we solved the problem by applying dropout and using more varied training data. These techniques forced the model to rely less on specific dataset artefacts and more on general patterns related to manipulation. Our findings are consistent with previous studies which show that supervised models can generalise reasonably well when trained on diverse datasets and supported with proper regularisation [5], [8]. Supervised models still face challenges when dealing with unknown deepfakes. However, our research highlights that success lies in the details. By carefully refining the model design and training setup, we can still achieve effective results.

B. The Failure of Anomaly Detection

The anomaly detection methods did not perform as expected. The autoencoder was unable to separate real from fake faces because the reconstruction error for both classes was almost identical. This suggests that the model was effectively "reconstructing everything," which made it unable to flag manipulated content. The VAE behaved differently but ultimately suffered from the same limitation. Even though it identified a threshold during validation, the model classified nearly every image as fake during testing. This resulted in perfect recall but poor precision. The issue seems to be that the deepfakes in FaceForensics++ and Celeb DF are visually close to real faces. Modern deepfakes are created using high-quality methods such as diffusion models and advanced identity preservation techniques [16], [35], which produce faces that still fall within the distribution learned by these models.

This matches trends reported in recent research. Reconstruction based detectors, especially simple autoencoders, tend to fail as deepfakes become more precise and lose the obvious pixel-level artefacts seen in earlier generations [2], [3]. As a result, basic anomaly detection is no longer adequate. Recent work, therefore, suggests moving towards more expressive generative models, feature-space anomaly scoring, or foundation-model-based detection [33], [36].

C. Analysis of Classifier Inconsistencies

Although the supervised classifier performed well overall, its accuracy varied across different manipulation styles. The model handled identity-swap methods such as FaceSwap with high confidence, likely because these methods introduce stronger spatial artefacts like blending boundaries or texture mismatches. In contrast, manipulations such as Neural Textures were more challenging. Since these methods change the expressions but keep the face structure intact, the visual clues are very faint [2]. This fits with earlier studies showing that spatial detectors usually miss these types of fakes. The artefacts are just too subtle and spread out in the fine textures to be caught easily [29]. Some studies have shown that temporal cues, including motion inconsistency and lip-speech

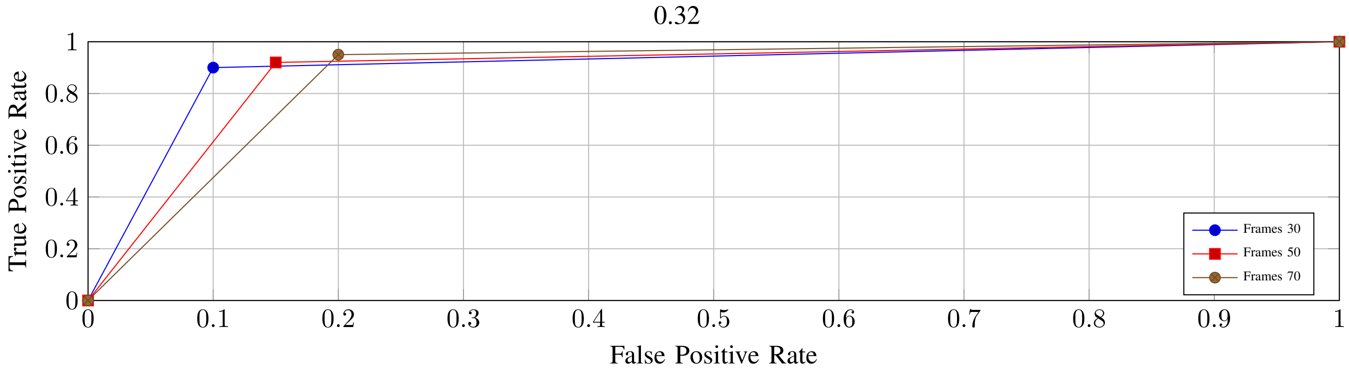

Fig. 7. FF++

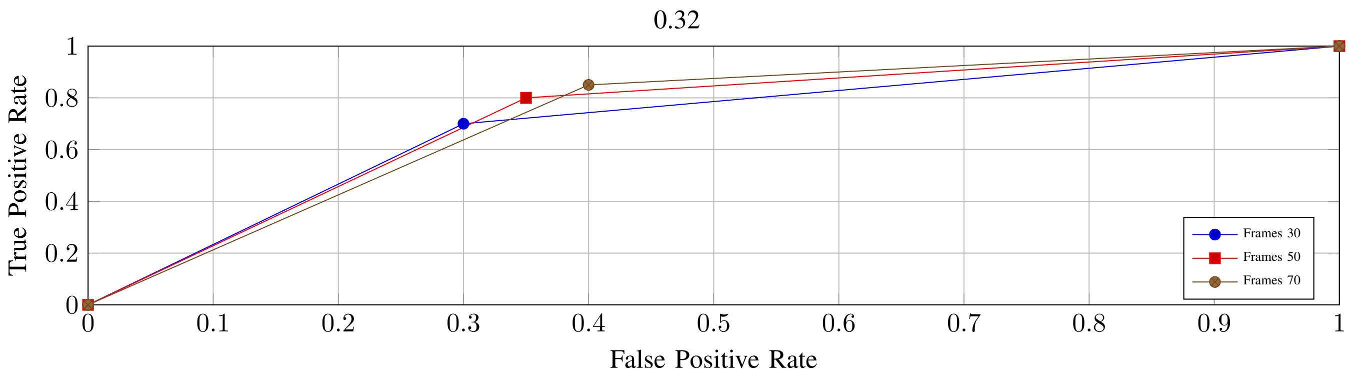

Fig. 8. Celeb-DF

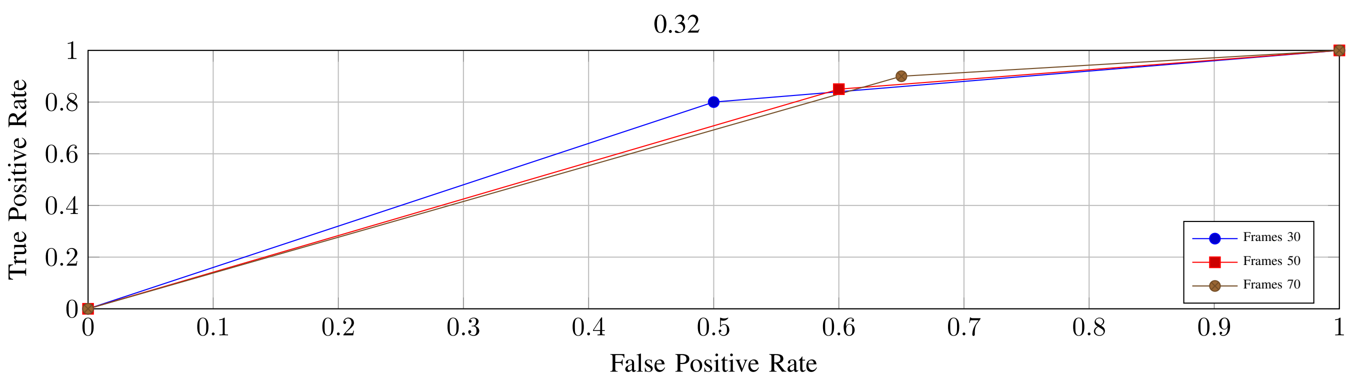

Fig. 9. DFDC

Fig. 10. ROC curves for different datasets under varying frame sampling conditions.

mismatch, can offer extra information that helps detect these kinds of forgeries [30], [31].

D. Limitations of This Study

This study has several limitations that are important to acknowledge. First, all three models were trained and tested on static frames. They do not use any temporal information, which is an important part of real deepfake videos. Many recent works show that temporal signals such as blink irregularities, inconsistent head motion, and lip-sync mismatches can significantly improve detection [31], [32].

Second, the anomaly detection models used here were intentionally simple. Autoencoders and VAEs are useful for studying basic reconstruction, but they lag behind today's advanced unsupervised models. Studies suggest that newer approaches such as contrastive self-supervision [20], latent space editing [37], and CLIP-based adapters [33] do a much better job at generalizing.

Third, although our dataset is diverse, it misses the full range of modern deepfakes. New tools are constantly appearing, including diffusion-based lip movements and multilingual

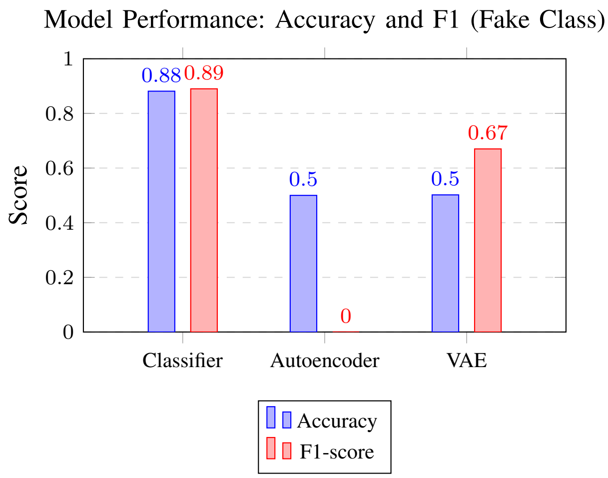

Fig. 11. Comparison of Accuracy and F1 (fake class) across models. Exact values are shown above bars.

synthetic speech [11], [38]. These create fresh challenges that aren't completely included in our data. These limitations make it clear that future work needs to explore temporal signals, stronger generative models, and larger cross-domain datasets to improve generalisation further.

VI. Conclusion

In this work, we compared a supervised XceptionNet classifier with two anomaly-based models, an Autoencoder and a VAE. Our aim was to understand whether unsupervised learning can generalise better to unseen deepfake styles. While it is generally understood that supervised models often outperform simple autoencoders, our results quantify this gap on modern high quality datasets. The extremely small difference in reconstruction error between real and fake samples shows that pixel based anomaly detection has reached its practical limits for current deepfake generation methods.

These findings suggest that simple reconstruction based anomaly detection is no longer sufficient for dealing with the deepfakes produced today. Modern generation techniques, including diffusion based face swapping [16], identity driven manipulation [39], and self consistent generative pipelines [40], create videos that are highly realistic. Because of this, reconstruction errors at the pixel level do not accurately reveal whether a face has been manipulated.This directly supports the current trend in deepfake forensics, where feature space anomaly detection using large vision encoders is replacing older pixel-level reconstruction methods.

Looking ahead, research needs to focus on methods that go far beyond basic autoencoders. Our findings reinforce the shift toward new, promising directions identified in recent work. The fact that both the autoencoder and VAE failed in pixel space tells us one thing: reliable detection models must move beyond raw reconstruction error and start operating in richer semantic feature spaces. Foundation models such as CLIP and DINOv3 can be adapted efficiently and have shown strong generalisation when used with parameter efficient tuning [33], [36]. There is also increasing evidence that combining spatial information with motion cues leads to better results, especially because many deepfakes contain very subtle temporal inconsistencies [30], [41]. Beyond that, researchers are finding value in augmenting the latent space or modelling style flows. We also see techniques that look at the latent space or style flows. Basically, these help the detector manage unfamiliar fakes [21], [37]. Self-supervised learning is another path people are taking, mostly because it lets the model learn without needing labelled samples [19]. Multimodal analysis is also gaining attention, particularly for spotting audiovisual deepfakes, which is done by looking for inconsistencies between speech and facial expressions. These methods could detect manipulations that single stream models might miss [42], [43].

As deepfake realism continues to advance, detection systems must become more flexible and less dependent on specific datasets. Our study shows that supervised learning still holds the advantage for current deepfake styles. Our results make this direction clearer. As deepfakes become even more realistic, it is clear that future detection systems cannot rely on pixel level reconstruction alone. The next generation of detectors will need to make use of stronger and more flexible feature representations learned from large foundation models, along with a better understanding of how faces move across time. In addition to this, combining information from both audio and visual signals will be important, since many modern manipulations remain visually convincing but reveal small inconsistencies when speech and facial expression are analysed together. Approaches that bring these elements together are much more likely to stay effective as deepfake generation continues to advance. This study quantifies how modern deepfakes affect pixel-space anomaly detection and provides implications for the shift toward foundation models.

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